

Extended-release formulations result in:

- A. Faster absorption
- B. Slower absorption
- C. Immediate peak
- D. No absorption

A drug is placed between the cheek and gum to facilitate direct absorption into the bloodstream without being swallowed. This method of administration is best described as:

- A. Sublingual administration
- B. Buccal administration
- C. Oral administration
- D. Rectal administration

A patient is given a drug formulation that is specifically designed to resist degradation in the acidic environment of the stomach and instead dissolve in the intestine. This formulation is referred to as:

- A. Immediate-release preparation
- B. Sustained-release preparation
- C. Enteric-coated preparation
- D. Intravenous formulation

A drug administered under the tongue produces rapid pharmacological effects because it avoids passage through the gastrointestinal tract and liver before entering systemic circulation. This advantage is known as:

- A. Increased bioavailability
- B. Avoidance of renal clearance
- C. Bypass of first-pass metabolism
- D. Enhanced gastric absorption

A critically ill patient requires a drug that must act immediately and be delivered with complete bioavailability. The most appropriate route of administration in this situation is:

- A. Oral route
- B. Intramuscular route
- C. Intravenous route
- D. Subcutaneous route

Drug administration via the rectal route may lead to inconsistent therapeutic effects because absorption through this route is:

- A. Complete and rapid
- B. Predictable and uniform
- C. Erratic and incomplete
- D. Independent of drug properties

A drug used in dental practice is known to be destroyed by gastric acid. The dentist prescribes an enteric-coated tablet. This formulation ensures that the drug is released in the:

- A. Stomach
- B. Oral cavity
- C. Small intestine
- D. Bloodstream

A dentist prescribes an extended-release analgesic to improve patient compliance after oral surgery. The main advantage of this formulation is:

- A. Rapid onset of action
- B. Frequent dosing required
- C. Prolonged duration of action
- D. Increased peak plasma levels

An intramuscular injection is given in a dental setting using a depot preparation. This type of formulation is characterized by:

- A. Immediate drug release
- B. Slow and sustained release
- C. No systemic effect
- D. Rapid elimination

A dentist avoids administering an irritating drug subcutaneously because it may cause:

- A. Increased absorption
- B. Hemorrhage
- C. Severe pain and necrosis
- D. Reduced metabolism

A patient with a respiratory condition is given a drug via inhalation. The advantage of this route is:

- A. Slow onset
- B. High first-pass metabolism
- C. Rapid drug delivery across a large surface area
- D. Erratic absorption

A patient who is vomiting after a dental procedure is given medication via the rectal route. One advantage of this route is:

- A. Complete avoidance of liver metabolism
- B. Predictable absorption
- C. Partial bypass of first-pass metabolism
- D. Immediate peak levels

Which of the following is a major disadvantage of the oral route?

- A. Pain
- B. Risk of infection
- C. First-pass metabolism
- D. Immediate effect

A drug is administered intravenously as a continuous infusion. Compared to IV bolus, this method:

- A. Produces higher peak levels
- B. Produces lower peak but sustained levels
- C. Delays absorption
- D. Causes first-pass metabolism

Which route is most appropriate for administering drugs in an unconscious patient where rapid action is required?

- A. Oral
- B. Rectal
- C. Intravenous
- D. Transdermal

A depot preparation administered intramuscularly results in prolonged drug action because:

- A. Drug is rapidly absorbed
- B. Drug is slowly released from injection site
- C. Drug bypasses liver
- D. Drug binds strongly to proteins